

Nil Patel

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Science in Computer Science and Electrical Engineering

Aug. 2024 – May. 2027

Coursework: Data Structures, Structure and Interpretation of Computer Programs, Computer Architecture, Foundations of Signals, Dynamical Systems and Information Processing,

EXPERIENCE

Software Engineering Intern

June 2025 – Aug. 2025

AsyncHealth

Berkeley, CA

- Engineered a local integration server using JavaScript, Node.js, and Express that leverages the **FHIR API** and PKCE-based **OAuth2** protocol to securely serialize/deserialize FHIR R4 JSON resources from **Epic's EHR** sandbox into the AsyncHealth web app, reducing manual data entry by **90%**
- Collaborated** with Dr. Pawan Jindal, CEO of Darena Health, to learn about FHIR protocol best practices, translating findings into actionable improvements for AsyncHealth's interoperability
- Built an embedded e-signature proof of concept to streamline patient onboarding—Node.js/Express + Axios backend to SignWell API

Software Engineering Intern

Jan. 2025 – June 2025

CloudIDR

Berkeley, CA

- Generated synthetic data using **CTGAN** in the form of instances of customers with unique and customarily distributed values such as CPU usage, Memory usage, etc
- Used Google Cloud Platform to extract **400,000+** rows of cloud computing data to train CTGAN model for data creation
- Prototyped an EC2 savings calculator using Python, that projected up to **40%** annual cost savings for enterprise customers
- Conducted Literature Review on several topic areas including synthetic data generation and AI model structures, "Performance Analysis of Machine Learning Centers Workload Prediction Models for Cloud", "MIT SuperCloud Dataset", "CTGAN Paper"

Product Management Intern

March 2023– July 2024

Virtual Medical Missions

Eastvale, CA

- Helped co-found a non-profit with Dr. French and Ms. Peraino to help rural villages in Godo, Kenya, receive medical access
- Led engineering team in creating a cost-effective vitals collection device that took heart rate, oxygen levels, etc
- Developed a clinical web app that captures patient vitals, signs, and symptoms, which generates a rule-based differential diagnosis for clinicians
- Presented** to Former CNUSD **Superintendent** Dr. Buenrostro along with CNUSD Board Members

PROJECTS

Build Your Own World | Java, Git, StdDraw

April 2025

- Developed an exploration game using **Java** which allows avatar to explore a randomly generated world
- Has **seed-driven** world generation which expands a world into a room graph, validates reachability and **synthesizes** corridor/hall geometry linking rooms producing deterministic worlds
- Built a modal encounter engine that swaps interaction interfaces when avatar engages with certain tiles, then restores exploration layer with original state applied

Pipelined 32-bit RISC-V CPU | Logisim

December 2025

- Built **data path** components including ALU, register file, branch comparator and immediate generator.
- Implemented** I-type, R-type, B-Type, load/store, jump, U-type instructions
- Wrote RISC-V **assembly** tests to validate CPU correctness and debug data path issues.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL/MySQL, JavaScript, HTML

Frameworks: React, Node.js

Developer Tools: Git, Visual Studio Code, PyCharm, IntelliJ